

The Formation of Marijuana Risk Perception in a Population of Substance Abusing Patients

Samuel T. Wilkinson¹ · Gerrit I. van Schalkwyk¹ ·
Larry Davidson¹ · Deepak C. D'Souza^{1,2,3}

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Abstract Risk perception has been shown to be protective with regard to marijuana use. Notably, the risk perception of marijuana in individuals with substance abuse problems varies significantly from that of the general public. Understanding how risk perception is formed in substance users could explain these differences and help predict the consequences of policy changes. Using this framework, we explored risk perception and its formation in a sample of substance abusing veterans. Semi-structured interviews were conducted with veterans who were receiving treatment for substance abuse. Interviews were recorded digitally, transcribed verbatim, and analyzed using inductive thematic analysis. A prominent perspective among the 31 participants was that marijuana is significantly different from other drugs because it is safe, not addictive, not associated with physical withdrawal, and has less overt behavioral effects than other substances. Many of these participants drew upon their own innocuous experiences with the drug in developing this perspective, more so than information from any other source. A contrasting narrative emphasized marijuana's capacity to cause negative social consequences, act as a gateway to the use of other, more harmful substances, and cause paranoia or worsen psychosis. In

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Samuel T. Wilkinson and Gerrit I. van Schalkwyk share first-authorship on this paper.

✉ Samuel T. Wilkinson
samuel.wilkinson@yale.edu

Gerrit I. van Schalkwyk
gerrit.vanschalkwyk@yale.edu

¹ Department of Psychiatry, Yale School of Medicine, New Haven, CT, USA

² Abraham Ribicoff Research Facilities, Connecticut Mental Health Center, New Haven, CT, USA

³ Schizophrenia and Neuropharmacology Research Group, VA Connecticut Healthcare System, West Haven, CT, USA

conclusion, individual experience with marijuana featured more prominently in informing risk perception than any other source of information. Our results and previous literature suggest that the significant disconnect between the individual experiences of substance users and the current clinical and legal policy towards marijuana may weaken the legitimacy of public policy or the authority of the medical community.

Keywords Marijuana · Marijuana abuse · Legalization · Substance abuse · Veterans · Drug policy

Introduction

Marijuana is a widely used substance that is currently undergoing substantial changes in its legal status. Results from the National Survey on Drug Use estimate that 17 million individuals in the US had used marijuana in the prior month [1]. Marijuana is now legal for medical purposes in 23 states, with recreational marijuana being legal in both Washington and Colorado. This rapidly evolving legal status points towards increasing acceptance for marijuana—a perspective echoed in the popular media, where marijuana legalization has often been framed as an attempt to provide chronically ill individuals access to a much-needed treatment [2]. By contrast, the academic literature has highlighted marijuana’s association with long-term health risks [3, 4], poor social outcomes [5, 6], and psychosis [7]. These potential risks highlight the importance of understanding how legalization may affect the use of marijuana.

A central concept in understanding the potential consequences of marijuana legalization is risk perception, most simply defined as the “perceptions of the negative effects from using drugs” [8]. Research has found that risk perception influences the usage of a drug in both adult and adolescent populations—specifically, a low risk perception is thought to relate to a higher pattern of use [9, 10]. Although recent data have shown that there continues to be a perception of substantial risk associated with marijuana use in the general population [11], other data indicate that risk perception is significantly lower in individuals with substance use disorders [12]. This finding supports previous work highlighting the importance of engaging the ‘affected community’, who may have different perspectives from the general population [13].

It has been hypothesized that legalization of marijuana may act to reduce risk perception, and thereby potentially increase use [14, 15]. Current epidemiological data in states where marijuana has recently been legalized for medical use does not support this idea [16], although it may be too early to detect real effects of policy change. There is also no comprehensive data describing the consequences of recreational legalization in the US. However, a better understanding of how risk perception is shaped may improve our ability to predict the consequences of legalization. It is particularly important that this be explored in individuals with substance use disorders. Empirical data showing that risk perception differs in this group suggests that it may be influenced by different factors than risk perception amongst the general population. By extension, the risk perception of substance users may be unique in the way it is affected by legalization.

In the current study, we explore risk perception towards marijuana in veterans with substance use disorders. In particular, we seek to better understand how risk perception might be formed in this group and why it might be different from the general population. In so doing we hope to further our current understanding of how legalization may affect risk perception in a particularly vulnerable population. Our study was deliberately designed to

provide important data that could be understood within this framework. To this end, we employed the use of qualitative methodology, allowing participants to provide open-ended narratives and to facilitate the exploration of underlying meanings and motivations attributed to specific perspectives [17]. Secondly, we purposively sampled individuals with substance use disorders in recognition of empirical and theoretical arguments supporting the relevance of obtaining the perspective of the ‘affected community’ [13]. Our study represents an early, yet significant description of these perspectives.

Methods

The study was approved by the Institutional Review Board of the West Haven, Connecticut Veterans Affairs Medical Center. Qualitative, semi-structured interviews were conducted with veterans who were receiving treatment for a substance use disorder at the West Haven, Connecticut Veterans Affairs Medical Center. The participants were residents in a program that included psycho-education, motivational interviewing and cognitive behavioral therapy, covering a range of substances. Potential participants were apprised of the study and were informed that their participation was voluntary. All of the 31 veterans who opted to participate completed the interview.

Interviews began with open-ended questions and invited participants to reflect on their risk perception of marijuana and how this may have been influenced by recent changes in its legal status. The primary questions were as follows:

- What things do you know about marijuana?
- Do you think it may be harmful? Why or why not?
- What do you think about marijuana legalization?
- How do you think this is going to change people’s attitudes about marijuana?

Participants were encouraged to expand on their initial responses to these questions. To this end, the interviewers employed a number of prompts such as enquiring about underlying reasons for certain views, or asking for further examples of specific experiences. Interviews were recorded digitally and transcribed. Data was analyzed (GVS) through the use of inductive thematic analysis: the data were subject to repeated rounds of coding during which themes were allowed to emerge ‘organically’—meaning that during this stage of analysis responses were provided descriptive labels imposing existing theoretical perspectives [17]. This, in combination with the use of open-ended questions, maximized our ability to give voice to the perspectives of our participants. At several stages of analysis, the process was reviewed by another investigator (STW) in order to improve validity. Themes were subsequently organized into hierarchies, and emphasis was given to themes that emerged consistently across the sample. Subsequently, themes were contextualized within existing theory. Interviewers had no role in the medical or psychiatric treatment of the research subjects. To facilitate the process of robust analysis, we used QSR Nvivo qualitative research software.

Results

Thirty-one veterans (30 male, 1 female), ranging in age from 24 to 65, completed interviews. All participants had a history of recent substance abuse and where engaged in active treatment. Although all patients had direct experience with marijuana use, the primary

substance of abuse varied across the sample. Participants in our sample provided a range of ideas about how they viewed marijuana. The most prominent idea was that marijuana was a low risk substance—a perspective informed by participants' own innocuous experiences with the substances. These participants were overwhelmingly in favor of marijuana being legalized. A minority opinion was that marijuana had profound potential for causing negative consequences, a perspective that was similarly informed by participants' own experiences with use.

Marijuana is Unlike Other Drugs

A dominant perspective in our group was that marijuana shared very few features with other drugs of abuse, and was considered to be a low-risk substance. As the following participant stated:

Well, there's no physical withdrawal. Maybe mental but I really don't find it to be, like, unsafe. I mean, especially for PTSD and anxiety, the best medication I've turned to was marijuana. It's much safer than a benzo and [other] medications they prescribed.... It helps with pain too. I don't find it to be like, something serious. I mean, yeah, it may do damage on your lungs and things like that but it's a lot safer than other street drugs and other medications that are, that's out there [Participant 17].

The above statement mentions a number of ideas that were prominent in our sample, including the sense that there was no withdrawal syndrome, that marijuana was helpful for specific psychiatric diagnoses, and that it was a lot safer than other street drugs. Other prominent ideas include the sense that it was not physically addictive, and that it could not be a drug because it was a natural substance. A further idea was that marijuana did not cause the same degree of behavioral effects as other drugs—said one participant *I never heard anybody who smoked marijuana just go off on somebody... going out robbing and fighting and stealing*—behaviors which this participant thought to be characteristic of other drugs of abuse. Many participants who had this perspective regarding marijuana expressed skepticism about people who did claim to have trouble with marijuana. Said one participant:

I also know a lot of people who have smoked a lot of marijuana and did not do so well. I think that resides in one's own motivation... (people) kind of use marijuana as a scapegoat for failures [Participant 2].

Some rejected the idea of it being addictive:

I think it's a false thing in their mind that they're addicted to marijuana. I mean, it's all in their head that they're addicted. I mean, an alcoholic could be addicted to alcohol and he can't stop and he's always in a detox center. Somebody who smokes marijuana might overdo it and keeps doing it on his own free will but he can stop and the marijuana is not addictive. So he can take a couple days off and then he'll be fine. I've done it. It's not hard to do. You just don't smoke it [Participant 8].

There was also skepticism that marijuana might be a gateway drug, with the same participant describing:

I don't believe it leads to other drugs... it's up to the individual if you want to go on and use other narcotics or drink. I just don't believe that [Participant 8].

Participants in this group compared marijuana to other drugs, emphasizing their sense that it was far less dangerous than many other substances—including legal substances such as alcohol. Participants further expressed skepticism about information they were receiving from clinicians about the risks of marijuana in terms of its addictive potential or role as a gateway drug. One participant even suggested that his clinician felt marijuana was safe, describing how *on a professional level, she can't really say it, but she does agree and she's a mental health clinician.*

Legalization an Obvious Choice

Consistent with this low risk conceptualization of marijuana was the sense amongst these participants that marijuana should be legalized. Regarding medical legalization, several participants felt that this was a way to provide a helpful treatment for a range of illnesses, and with fewer side effects:

With medical marijuana, definitely [should be legalized]. I believe that, I think some of the medications or the different things that they use with certain illnesses, I think that the marijuana has less, I know it has less side effects and your body I think accepts it easier without all the side effects [Participant 16].

Specific examples included opiate pain medications, with patients arguing that the *benefit is greater on marijuana than it is on Oxycodone and there's less addiction.* Regarding legalization for recreational purposes, a common theme that emerged was that this would make marijuana more acceptable, and more widely utilized:

I think it'll increase, it will definitely increase, cause I think there's a lot of closet users, there's a lot of people who hide it because it's illegal and then there's a lot of people who just don't do it because it's illegal. Or it's just not, they don't know how to get it [Participant 4].

Participants described how legalization would lead to people being less likely to *'frown upon it so much'*, and that it would lead to marijuana being promoted as openly as alcohol. However, many participants did not consider this to be a negative consequence of legalization, as they considered marijuana to be less dangerous than alcohol—a legal substance—and secondly, felt that it was inappropriate for marijuana users to be considered criminals. As an extension of the latter idea was found in the descriptions of ways in which recreational legalization would reduce the criminal activity associated with selling marijuana:

I believe that it should be legal for people of a certain age, 21, regulated by the government because in a case I compare it to alcohol. There's gotta be some regulation or we're gonna have more of the gangs and all that kind of crap that goes on right now behind marijuana. The criminal element is using it so much to fund stuff and to guns and violence and everything [Participant 18].

There was a sense that legalization would *'run the cartels out'* and would make it safer for people to procure marijuana because somebody *'just looking to get high doesn't have to go to a dangerous place'*. Relatedly, by breaking the connection to drug dealers, it was also felt that legalization would reduce the capacity for marijuana to act as a gateway drug—an assertion premised on the idea that individuals in our participants' communities were first offered more serious drugs by dealers from whom they were purchasing marijuana. Summatively, there was a strong sense amongst participants that marijuana legalization was both appropriate and potentially of significant benefit.

A Contrasting Experience

A less prominent and contrasting perspective was present, and appeared to be informed by some participants' own negative experiences with marijuana. The below quote is particularly illustrative:

In different periods of my life it affected me differently. This most recent period it caused tremendous relationship problems with my fiancée, caused financial problems cause I was spending a lot of money on it, caused me to decrease my performance at work. I've been arrested before for possession of marijuana.... I jeopardized my military status.... And then got horrible grades in college because of it so a lot of negative consequences [Participant 15].

This same participant described a strong sense that marijuana was physically addictive, and other participants stated that, although they could not imagine becoming 'physically addicted' to marijuana, 'psychologically—now there's a whole new ball of wax... you crave the feeling it gives you'. Consistent with this more serious conceptualization was the sense that marijuana could, in fact have serious behavior effects, particularly in individuals with psychiatric illness:

I could remember in the past when I smoked with people they either became paranoid or, you know, get the munchies... so it's definitely a serious mood changer....

And later,

I suffer from [psychotic illness] and so I'm already somewhat paranoid and on edge and marijuana seems to take me right over the edge [Participant 9].

These individuals also described how they had experienced marijuana as being a gateway drug, saying *Yeah, it does lower your inhibition and I did do other drugs and usually after I smoked a joint*. Another participant described marijuana as *opening the floodgates* to harder narcotics. Participants who shared this view cited the concept of 'lowered inhibitions' as a common way in which marijuana could act as a gateway drug. Another mechanism of interest was the sense that owing to marijuana being illegal, it led to users coming into contact with drug dealers, thus acting as a gateway to further drug abuse.

Of note, even in this group participants had mixed feelings regarding legalization. Regarding medical legalization, a few participants expressed some concern that patients might try and fabricate symptoms in order to receive prescriptions for marijuana, but at the same time argued that this was unlikely to be a significant phenomenon given how widely available marijuana was within their community. Furthermore some participants felt that recreational legalization would have negative consequences, including poorer workplace performance and more widespread use. Others were less against legalization but pointed out how legalization would require the development of regulations in a similar way to alcohol.

Discussion

Our results provide a number of findings of particular relevance. Firstly, participants in our study were divided in terms of the risk they attributed to marijuana. One perspective was that marijuana was significantly different than other drugs, being non-addictive, having no withdrawal syndrome, and causing relatively limited behavioral effects. The contrasting

narrative conceptualized marijuana as an addictive substance with significant social consequences. Individuals who attributed a higher risk perception to marijuana were those who described direct experience of its negative effects. Conversely, those whose experiences with marijuana were comparatively innocuous exhibited a lower perception of risk (see Table 1). Hence, in this substance using population, personal experience may contribute more to the development of risk perception than information from clinicians or the popular media. Indeed, every member of our sample was engaged in a program where the risks of marijuana were clearly described, including evidence of dependence and a withdrawal syndrome after cessation of use [18–20]. It is helpful to consider this finding from a motivational interviewing perspective, where strategies such as ‘looking back’ and ‘looking forward’ may elicit reflections on the past and future consequences of a specific behavior [21]. Individuals who cannot identify any losses associated with their use of marijuana may find no discrepancy between their current beliefs and experiences with marijuana, making them resistant to considering other perspectives. It is plausible that this phenomenon contributes to the wide variation in risk perception seen in our sample and in previous literature [10].

Given the low risk perception of many of our participants, it is unsurprising that the prominent trend in our sample was for participants to support marijuana legalization for recreational purposes. Participants drew comparisons to other legal substances to which they ascribed greater risk, and felt that this was a prominent reason for marijuana to be legalized. This finding resonates with previous literature which notes that, in many instances, laws and practice regarding marijuana have become highly divergent [22]. For example, though most individuals who use cannabis do not experience adverse effects, the drug’s illegal status suggests that all use is harmful [23]. This discrepancy between individual experience and public policy has been thought to weaken the validity of laws intending to protect public health [22], particularly when some users report subjective health benefits from marijuana use [24]. In our sample, many participants had innocuous experiences with marijuana, which were different with those they attributed to other illegal substances. Participants were so confident in these views that they expressed disbelief in the narratives of individuals who did describe more negative experiences. It is therefore possible that the current illegal status of marijuana impairs the legitimacy of efforts aimed at increasing awareness of the risks of marijuana—the divide between law and practice is so great that participants reject any message of risk, instead adopting a somewhat extreme “low-risk” perspective that is more consistent with their own experiences.

In addition, many participants felt that legalization of recreational marijuana would act to move it out of the ‘drug space’, thereby making the process of procurement safer, weakening drug cartels, and reducing its capacity to act as a gateway drug. The later assertion has been previously described, and is consistent with a hypothesis of ‘differential association’, which posits that the primary way through which marijuana use leads to other drug use is through the association of marijuana users with drug culture and dealers [25]. If marijuana could be obtained legally, this association would be weakened and hence the drug’s role as a gateway diminished.

In the case of medical legalization, we again see how comparative risk perception is informed by experience, with participants drawing comparisons between marijuana (with which they had experienced little adverse consequences) and other drugs (with which many reported direct or indirect experience of harmful effects). It is important to note that by framing marijuana for medical purposes as a legal issue, little attention is paid to whether or not a burden of proof is met in terms of it being an evidence-based treatment. Indeed, as mentioned previously, a recent meta-analysis suggests that further evidence is required to substantiate the use of marijuana for pain, especially in light of frequent

Table 1 Personal experience with marijuana informs risk perception in a population of veterans with substance abuse

Majority opinion Innocuous experiences with marijuana inform a lower perception of risk	Minority opinion Harmful experiences with marijuana inform a higher perception of risk
Well you can just crave the feeling that it gives you. A feeling of well-being, a feeling of calmness. It's good if you don't have any appetite. Things like that. I see more beneficial uses of it than I do see [harms]. I don't feel like it's a gateway drug to heroin, to meth or anything like that [Participant 18]	It does lower your inhibition and I did do other drugs and usually after I smoked a joint [Participant 10]
I don't think it's addictive, such as other drugs that I've experienced [Participant 4]	Long term use, I think [is harmful]. I mean, if you've ever smoked a joint you can see the residue, the resin, the tar, the stuff that it leaves behind if you have a nice big, fat joint. That stuff is leaving resin and residues and things of that nature on your, in your lungs [Participant 9]
I don't think [marijuana is harmful]. I know I used to smoke weed way back when I was younger and the only thing marijuana did to me was ... getting hungry and make me wanna eat [Participant 11]	In different periods of my life it affected me differently. This most recent period it caused tremendous relationship problems with my fiancée, caused financial problems cause I was spending a lot of money on it, caused me to decrease my performance at work. I've been arrested before for possession of marijuana.... I jeopardized my military status.... And then got horrible grades in college because of it so a lot of negative consequences [Participant 15]
I think it's a false thing in their mind that they're addicted to marijuana. I mean, it's all in their head that they're addicted.... Somebody who smokes marijuana might overdo it and keeps doing it on his own free will but he can stop and the marijuana is not addictive. So he can take a couple days off and then he'll be fine. I've done it. It's not hard to do. You just don't smoke it [Participant 12]	It alters your mental status in a significant way. It seems like there's a big push to make marijuana seem like it's not a big deal and it's very socially acceptable like in media, pop culture, things like that.... I think for everyone it's a drug, it's just some people it affects differently than other people... and that's kind of like ... playing with fire [Participant 15]
I think it's different than other drugs and I mean, it's I never heard nobody really who smoke marijuana just go off on somebody or going out there and just hurting somebody, per say, you know versus other drugs that people take, you know [Participant 6]	

adverse events reported [26]. Furthermore, there are no randomized controlled trials for the use of marijuana in many other conditions for which it is now legal, including Posttraumatic Stress Disorder, Alzheimer's Disease, and Parkinson's Disease. However, when the issue is one of legality, the discourse of substance abusers shifts towards comparative assessments of harm, which are informed by personal experience with the drug.

The need to facilitate a legitimate discourse about the risks of marijuana is of crucial importance—many participants felt that marijuana was a low risk substance, and that individuals who were having challenges were mistaken in attributing their difficulties to marijuana. On the other hand, a significant minority of participants provided detailed narratives of profound social consequences associated with their marijuana use. Their narratives are supported by existing scientific data documenting a number of adverse effects of marijuana, including lower IQ, poor school performance, and risks of motor vehicle accidents [3]. Further, tolerance, addiction, and withdrawal associated with marijuana are well described [18, 20, 27–29], as is the drug's association with immediate and transient psychosis as well as permanent psychotic disorders [7, 30]. Hence, individuals who develop marijuana use disorders may struggle to have their problems regarded as legitimate by their peers. This is particularly relevant in our population, as much of the substance abuse treatment available to veterans relies on group therapy. Importantly, it should be highlighted

that although legalization may act to reduce barriers to balanced discourse on risk, this does not imply that such discourse will take place. Therefore, it is likely that for legalization to be effective in this regard, it would need to be accompanied by a balanced strategy to deliver a balanced message about the real risks associated with marijuana use.

This study serves to expand our understanding of the current risk perception towards marijuana in veterans with substance use disorders and how this may be influenced in the context of marijuana's evolving legal status. In particular, we offer a possible explanation for the finding that risk perception differs in individuals with substance users versus the general population. Specifically, our data suggest that individual experience with the effects of marijuana may contribute more meaningfully to the development of risk perception than information from any other source, including medical professionals and the implicit message of its legal status. Notably, our findings echo previous the work of Duff et al. [22] and Temple et al. [23] providing narrative data supporting the idea that when 'policy' is highly divergent from 'practice', it may lose legitimacy. We extend this theory in two ways. Firstly, in our sample, experiences were not only different from legal policy, but also contrasted with the clinical content being communicated to participants in their substance use treatment. Secondly, we found that in addition to this divergence leading to a loss of legitimacy, it had the effect of pushing participants towards a position where they were prone to invalidate the contrasting experiences of individuals who had in fact experienced challenges in the context of marijuana use.

The primary weakness of our study was the fact that we were limited in our capacity to obtain detailed demographic information from our participants. In addition, as our study was conducted in veterans, our findings may not be generalizable to other groups, including adolescents and veterans without substance abuse problems.

In summary, many of our participants described experiences and beliefs regarding marijuana that distinguished it as being of much lower risk than other drugs. These beliefs and experiences appeared to have made participants resistant to internalizing information about the risks of marijuana, despite receiving significant education in this regard. It would appear that there is a significant disconnect between the experiences of substance users and the current clinical and legal policy towards marijuana. Unfortunately, in the context of this disconnect, participants who had not experienced adverse effects associated with marijuana often failed to recognize that some individuals could suffer serious negative consequences. In considering future public policy and legal changes, it is relevant to consider the role of personal narratives and motivations in guiding the behaviors of individuals with substance use disorders. Doing so may help produce a more functional discourse that allows for a balanced presentation of the risks of marijuana, and improve our capacity to predict the consequences of legal changes.

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Conflict of interest None.

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Samuel T. Wilkinson, MD studied Mechanical Engineering (minor in Chemistry) at Brigham Young University. Afterwards, he obtained his medical degree from Johns Hopkins University School of Medicine in Baltimore. During his medical school training, he conducted research in Obsessive–Compulsive Disorder in people with developmental disabilities and earned the Doak Walker Award in Psychiatry for his research efforts. Upon graduation, he was awarded the Samuel Novey Prize in Psychological Medicine. At Yale, where he is part of the Neuroscience Research Training Program, he has developed an interest in cannabinoid-psychosis interactions as well as in new therapies for treatment-resistant depression.

Gerrit I. van Schalkwyk, MB, ChB is a PGY-3 adult psychiatry resident in the Yale University Department of Psychiatry. Gerrit has a particular interest in early adulthood, with clinical work and research experience exploring mental illness in this population. In particular, Gerrit is interested in the experiences of young adults with Autism Spectrum Disorders, psychotic disorders, and the complex intersection between these conditions. In addition, Gerrit has interests in research mentorship, and LGTBQ issues related to psychiatry and mental health.

Larry Davidson, PhD is involved in research and consultation/training work focuses on processes of recovery from and in serious mental illnesses and addictions and the development and evaluation of innovative policies and programs to promote the recovery and community inclusion of individuals with these conditions. We utilize mixed qualitative and quantitative methods, and typically conduct our research in a participatory and inclusive fashion that involves people in recovery in all aspects and phases of the research process.

Deepak C. D'Souza, MD, MBBS is Professor of Psychiatry, Yale University School of Medicine and VA Connecticut Healthcare System. Dr. D'Souza is an active clinician, teacher and researcher. Dr. D'Souza's research involves the use of drug probes to evaluate the contributions of various neurotransmitter systems to the pathophysiology of psychotic states and cognitive function, as well as laboratory models to test therapies for schizophrenia, and clinical trials for schizophrenia and cannabis dependence. He is funded by NIDA, NIMH, NIAAA, the VA and several foundations.